

Exercise 19.3

1) Evaluate the following:

$$a) \int \frac{1}{\sqrt{9-x^2}} dx$$

$$= \int \frac{1}{\sqrt{3^2-x^2}} dx$$

$$\text{let } x = 3 \sin \theta \Rightarrow \theta = \sin^{-1}\left(\frac{x}{3}\right) \\ dx = 3 \cos \theta d\theta$$

Now,

$$\int (3^2-x^2)^{-1/2} dx$$

$$= \int \frac{1}{\sqrt{3^2-(3\sin\theta)^2}} dx$$

$$= \int \frac{1}{\sqrt{9-9\sin^2\theta}} dx$$

$$= \int \frac{1}{\sqrt{9(1-\sin^2\theta)}} dx$$

$$= \int \frac{1}{3 \cos \theta} dx$$

$$= \int \frac{1}{\cancel{3 \cos \theta}} \times \cancel{3 \cos \theta} d\theta$$

$$= \int 1 d\theta$$

$$= \theta + C$$

$$= \sin^{-1} \frac{x}{3} + C$$

$$b) \int \frac{1}{x^2 \sqrt{4-x^2}} dx$$

$$\text{let } x = 2 \sin \theta \Rightarrow \sin \theta = \frac{x}{2} = \frac{p}{h}$$

$$\frac{dx}{d\theta} = 2 \cos \theta$$

$$b = \sqrt{4-x^2}$$

$$dx = 2 \cos \theta d\theta$$

Now,

$$\int \frac{1}{(2 \sin \theta)^2 \sqrt{4-4 \sin^2 \theta}} \times 2 \cos \theta d\theta$$

$$= \int \frac{1}{4 \sin^2 \theta \sqrt{4(1-\sin^2 \theta)}} \times 2 \cos \theta d\theta$$

$$= \int \frac{1}{4 \sin^2 \theta \times 2 \cos \theta} \times 2 \cos \theta d\theta$$

$$= \int \frac{1}{4 \sin^2 \theta} d\theta$$

$$= \frac{1}{4} \int \frac{1}{2 \sin^2 \theta} d\theta$$

$$= \frac{1}{4} \int \frac{1}{\cos^2 \theta} d\theta = \frac{1}{4} \int \sec^2 \theta d\theta$$

$$= \frac{1}{4} (-\cot \theta + C)$$

$$= \frac{-\sqrt{4-x^2}}{4x} + C$$

$$c) \int \frac{x^2}{\sqrt{9-x^2}} dx$$

$$\rightarrow \text{let } x = 3 \sin \theta \Rightarrow \theta = \sin^{-1}\left(\frac{x}{3}\right)$$

$$\frac{dx}{d\theta} = 3 \cos \theta$$

$$b = \sqrt{9-x^2}$$

$$dx = 3 \cos \theta d\theta$$

Now,

$$\int \frac{x^2}{\sqrt{9-x^2}} dx$$

$$= \int \frac{(3 \sin \theta)^2}{\sqrt{9-9 \sin^2 \theta}} \times 3 \cos \theta d\theta$$

$$= \int \frac{9 \sin^2 \theta}{3 \cos \theta} \times \cancel{3 \cos \theta} d\theta$$

$$= 9 \int \sin^2 \theta d\theta$$

$$= \frac{9}{2} \int 2 \sin^2 \theta d\theta$$

$$= \frac{9}{2} \int (-\cos 2\theta + 1) d\theta$$

$$= \frac{9}{2} \left[\int -\cos 2\theta d\theta + \int 1 d\theta \right]$$

$$= \frac{9}{2} \left[\frac{-\sin 2\theta}{2} + \theta \right] + C$$

$$= \frac{9(-\sin 2\theta)}{2} + \frac{9\theta}{2} + C$$

$$= -\frac{9}{2} \times \cancel{2} \sin \theta \times \cos \theta + \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) + C$$

$$= -\frac{9}{2} \times \frac{x}{3} \times \frac{\sqrt{9-x^2}}{3} + \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) + C$$

$$= \frac{9}{2} \sin^{-1}\left(\frac{x}{3}\right) - \frac{1}{2} x \sqrt{9-x^2} + C$$

2) Evaluate the following:

a) $\int \frac{1}{x^2 \sqrt{1+x^2}} dx$

let $x = \tan \theta \Rightarrow \tan \theta = \frac{x}{1} = \frac{p}{b}$

$\frac{dx}{d\theta} = \sec^2 \theta$

$h = \sqrt{1+x^2}$

$dx = \sec^2 \theta d\theta$

Now,

$\int \frac{1}{\tan^2 \theta \sqrt{1+\tan^2 \theta}} \times \sec^2 \theta d\theta$

$= \int \frac{1}{\tan^2 \theta \times \sec \theta} \times \sec^2 \theta d\theta$

$= \int \frac{\sec \theta}{\tan^2 \theta} d\theta$

$= \int \frac{\cos^2 \theta}{\sin^2 \theta} \times \frac{1}{\cos \theta} d\theta$

$= \int \frac{\cos \theta}{\sin \theta} \times \frac{1}{\sin \theta} d\theta$

$= \int \cot \theta \cdot \operatorname{cosec} \theta d\theta$

$= -\operatorname{cosec} \theta + C$

$= -\frac{\sqrt{1+x^2}}{x} + C$

b) $\int \frac{1}{x \sqrt{x^2+9}} dx$

let $x = 3 \tan \theta \Rightarrow \tan \theta = \frac{x}{3} = \frac{p}{b}$

$\frac{dx}{d\theta} = 3 \sec^2 \theta$

$h = \sqrt{x^2+9}$

$dx = 3 \sec^2 \theta d\theta$

Now,

$$\int \frac{1}{3 \tan \theta \sqrt{9 \tan^2 \theta + 9}} \times 3 \sec^2 \theta d\theta$$

$$= \int \frac{1}{3 \tan \theta \times \cancel{3} \sec \theta} \times \cancel{3} \sec^2 \theta d\theta$$

$$= \int \frac{1}{3} \times \frac{\cos \theta}{\sin \theta} \times \frac{1}{\cos \theta} d\theta$$

$$= \frac{1}{3} \int \operatorname{cosec} \theta d\theta$$

$$= \frac{1}{3} (-\ln |\operatorname{cosec} \theta + \cot \theta|) + C$$

$$= -\frac{1}{3} \ln |\operatorname{cosec} \theta + \cot \theta| + C$$

$$= -\frac{1}{3} \ln \left| \frac{\sqrt{x^2+9}}{x} + \frac{3}{x} \right| + C$$

$$= -\frac{1}{3} \ln \left| \frac{3 + \sqrt{x^2+9}}{x} \right| + C$$

c) $\int \frac{1}{(a^2+x^2)^2} dx$

let $x = a \tan \theta \Rightarrow \theta = \tan^{-1} \left(\frac{x}{a} \right) = \frac{\theta}{b}$

$$\frac{dx}{d\theta} = a \sec^2 \theta$$

$$h = \sqrt{x^2 + a^2}$$

$$dx = a \sec^2 \theta d\theta$$

Now,

$$\int \frac{1}{(a^2 + a^2 \tan^2 \theta)^2} \times a \sec^2 \theta d\theta$$

$$= \int \frac{1}{(a^2 \sec^2 \theta)^2} \times a \sec^2 \theta d\theta$$

$$= \int \frac{1}{a^2 \sec^2 \theta \times a^2 \sec^2 \theta} \times a \sec^2 \theta d\theta$$

$$= \int \frac{1}{a^3 \sec^2 \theta} d\theta$$

$$= \frac{1}{a^3} \int \cos^2 \theta d\theta$$

$$= \frac{1}{2a^3} \int 2 \cos^2 \theta d\theta$$

$$= \frac{1}{2a^3} \int (1 + \cos 2\theta) d\theta$$

$$= \frac{1}{2a^3} \left[\int 1 d\theta + \int \cos 2\theta d\theta \right]$$

$$= \frac{1}{2a^3} [\theta + \sin 2\theta] + C$$

$$= \frac{1}{2a^3} \left[\tan^{-1} \frac{x}{a} + 2 \sin \theta \cdot \cos \theta \right] + C$$

$$= \frac{1}{2a^3} \tan^{-1} \left(\frac{x}{a} \right) + \frac{1}{a^3} \times \cancel{x} \times \frac{x}{\sqrt{x^2 + a^2}} \times \frac{a}{\sqrt{x^2 + a^2}} + C$$

$$= \frac{1}{2a^3} \tan^{-1} \left(\frac{x}{a} \right) + \frac{1}{a^3} \left[\frac{ax}{x^2 + a^2} \right] + C$$

3) Evaluate the following:

a) $\int \frac{\sqrt{x^2 - 4}}{x} dx$

let $x = 2 \sec \theta \Rightarrow \sec \theta = \frac{x}{2} = \frac{h}{b} \Rightarrow p = \sqrt{x^2 - 4}$

$$\frac{dx}{d\theta} = 2 \sec \theta \cdot \tan \theta$$

$$dx = 2 \sec \theta \cdot \tan \theta d\theta$$

Now,

$$\int \frac{\sqrt{4 \sec^2 \theta - 4}}{\cancel{x} \sec \theta} \times \cancel{x} \sec \theta \cdot \tan \theta d\theta$$

$$\begin{aligned}
 & \int \frac{\sqrt{4(\sec^2 \theta - 1)}}{1} \times \tan \theta \, d\theta \\
 &= \int 2 \tan \theta \times \tan \theta \, d\theta \\
 &= \int 2 \tan^2 \theta \, d\theta \\
 &= 2 \int (\sec^2 \theta - 1) \, d\theta \\
 &= 2 [\int \sec^2 \theta \, d\theta - \int d\theta] \\
 &= 2 [\tan \theta - \theta] + C \\
 &= 2 \left[\frac{\sqrt{x^2 - 4}}{2} - \sec^{-1} \left(\frac{x}{2} \right) \right] + C \\
 &= \sqrt{x^2 - 4} - \sec^{-1} \frac{x}{2} + C
 \end{aligned}$$

b) $\int \frac{L}{(x^2 - a^2)^{3/2}} \, dx$

→ let $x = a \sec \theta \Rightarrow \sec \theta = \frac{x}{a} = \frac{h}{b} \Rightarrow P = \sqrt{x^2 - a^2}$
 $\frac{dx}{d\theta} = a \sec \theta \cdot \tan \theta$
 $dx = a \sec \theta \cdot \tan \theta \, d\theta$

Now,

$$\begin{aligned}
 & \int \frac{1}{(a^2 \sec^2 \theta - a^2)^{3/2}} \times a \sec \theta \cdot \tan \theta \cdot d\theta \\
 &= \int \frac{1}{[a^2 (\sec^2 \theta - 1)]^{3/2}} \times a \sec \theta \cdot \tan \theta \cdot d\theta \\
 &= \int \frac{1}{a^3 \cdot \tan^3 \theta} \times a \sec \theta \cdot \tan \theta \, d\theta \\
 &= \int \frac{1}{a^2 \tan^2 \theta} \times \sec \theta \, d\theta \\
 &= \frac{1}{a^2} \int \frac{\cancel{\cos \theta}}{\sin^2 \theta} \times \frac{\cos \theta}{\cancel{\cos \theta}} \, d\theta
 \end{aligned}$$

$$= \frac{1}{a^2} \int \operatorname{cosec} \theta \cdot \cot \theta \, d\theta$$

$$= \frac{1}{a^2} (-\operatorname{cosec} \theta) + C$$

$$= -\frac{1}{a^2} \operatorname{cosec} \theta + C$$

$$= -\frac{1}{a^2} \frac{a}{\sqrt{x^2 - a^2}} + C$$

$$= -\frac{x}{a^2 \sqrt{x^2 - a^2}} + C$$

c) $\int \frac{1}{x^2 \sqrt{x^2 - 4}} \, dx$

let $x = 2 \sec \theta \Rightarrow \sec \theta = \frac{x}{2} = \frac{h}{b} \Rightarrow p = \sqrt{x^2 - 4}$

$$\frac{dx}{d\theta} = 2 \sec \theta \cdot \tan \theta$$

$$dx = 2 \sec \theta \cdot \tan \theta \cdot d\theta$$

Now,

$$\int \frac{1}{4 \sec^2 \theta \sqrt{4 \sec^2 \theta - 4}} \times 2 \sec \theta \cdot \tan \theta \cdot d\theta$$

$$= \int \frac{1}{4 \sec \theta \times 2 \tan \theta} \times 2 \sec \theta \cdot \tan \theta \cdot d\theta$$

$$= \frac{1}{4} \int \cos \theta \, d\theta$$

$$= \frac{1}{4} \sin \theta + C$$

$$= \frac{1}{4} \frac{\sqrt{x^2 - 4}}{x} + C$$

$$= \frac{\sqrt{x^2 - 4}}{4x} + C$$

4) Integrate:

$$a) \int \sqrt{\frac{x}{a-x}} dx$$

$$\text{Here, } \sqrt{a-x} = \sqrt{(\sqrt{a})^2 - (\sqrt{x})^2}$$

$$\text{let } \sqrt{x} = \sqrt{a} \sin \theta \Rightarrow \sin \theta = \frac{\sqrt{x}}{\sqrt{a}} = \frac{p}{h}$$

$$x = a \sin^2 \theta$$

$$b = \sqrt{a-x}$$

$$\frac{dx}{d\theta} = a \frac{d \sin^2 \theta}{d \sin \theta} \times \frac{d \sin \theta}{d \theta}$$

$$= a \times 2 \sin \theta \cdot \cos \theta$$

$$dx = 2a \sin \theta \cdot \cos \theta \cdot d\theta$$

Now,

$$\int \sqrt{\frac{a \sin^2 \theta}{a - a \sin^2 \theta}} \times 2a \sin \theta \cdot \cos \theta \cdot d\theta$$

$$= \int \sqrt{\frac{a \sin^2 \theta}{a \cos^2 \theta}} \times 2a \sin \theta \cdot \cos \theta \cdot d\theta$$

$$= \int \frac{\sin \theta}{\cos \theta} \times 2a \sin \theta \cdot \cos \theta \cdot d\theta$$

$$= \int 2a \sin^2 \theta d\theta$$

$$= a \int 2 \sin^2 \theta d\theta$$

$$= a \int (1 - \cos 2\theta) d\theta$$

$$= a \left[\int 1 d\theta - \int \cos 2\theta d\theta \right]$$

$$= a \left[\theta - \frac{\sin 2\theta}{2} \right] + c$$

$$= a\theta - \frac{a \sin 2\theta}{2} + c$$

$$= a \sin^{-1} \left(\frac{\sqrt{x}}{\sqrt{a}} \right) - \frac{a \times 2 \sin \theta \cdot \cos \theta}{2} + c$$

$$= a \sin^{-1}\left(\sqrt{\frac{n}{a}}\right) - a \times \frac{\sqrt{n}}{\sqrt{a}} \times \frac{\sqrt{a-n}}{\sqrt{a}} + c$$

$$= a \sin^{-1}\sqrt{\frac{n}{a}} - \sqrt{n} \cdot \sqrt{a-n} + c$$

b) $\int \sqrt{\frac{a-n}{a+n}} dn$

$$= \int \sqrt{\frac{a-n}{a+n}} \times \frac{a-n}{a-n} dn$$

$$= \int \sqrt{\frac{(a-n)^2}{a^2-n^2}}$$

$$= \int \frac{a-n}{\sqrt{a^2-n^2}}$$

let $n = a \sin \theta \Rightarrow \sin \theta = \frac{n}{a} = \frac{p}{h}$

$$\frac{dn}{d\theta} = a \cos \theta$$

$$dn = a \cos \theta d\theta$$

$$b = \sqrt{a^2 - n^2}$$

Now,

$$\int \frac{a - a \sin \theta}{\sqrt{a^2 - a^2 \sin^2 \theta}} \times a \cos \theta d\theta$$

$$= \int \frac{a(1 - \sin \theta)}{a \cos \theta} \times a \cos \theta d\theta$$

$$= a \int (1 - \sin \theta) d\theta$$

$$= a [\int 1 d\theta - \int \sin \theta d\theta]$$

$$= a [\theta - (-\cos \theta)] + c$$

$$= a \left[\sin^{-1}\left(\frac{n}{a}\right) + \frac{\sqrt{a^2 - n^2}}{a} \right] + c$$

$$= a \sin^{-1} \frac{n}{a} + \sqrt{a^2 - n^2} + c$$

5) Evaluate the following definite integrals.

$a) \int_0^a \sqrt{a^2 - x^2} dx$	When $x=0$ $\theta=0$ When $x=a$ $\theta = \frac{\pi}{2}$
let $x = a \sin \theta$ $\frac{dx}{d\theta} = a \cos \theta$ $dx = a \cos \theta d\theta$	

Now,

$$\int_0^{\pi/2} \sqrt{a^2 - a^2 \sin^2 \theta} \times a \cos \theta d\theta$$

$$= \int_0^{\pi/2} (a \cos \theta \times a \cos \theta) d\theta$$

$$= \int_0^{\pi/2} a^2 \cos^2 \theta d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} 2 \cos^2 \theta d\theta$$

$$= \frac{a^2}{2} \int_0^{\pi/2} (\cos 2\theta + 1) d\theta$$

$$= \frac{a^2}{2} \left[\int_0^{\pi/2} \cos 2\theta d\theta + \int_0^{\pi/2} 1 d\theta \right]$$

$$= \frac{a^2}{2} [\sin 2\theta + \theta]_0^{\pi/2}$$

$$= \frac{a^2}{2} \left[\sin \pi + \frac{\pi}{2} - \sin 0 - 0 \right]$$

$$= \frac{a^2}{2} \left[\sin \pi + \frac{\pi}{2} \right]$$

$$= \frac{a^2}{2} \left[0 + \frac{\pi}{2} \right]$$

$$= \frac{\pi a^2}{4}$$

$$b) \int_1^{\sqrt{2}} \frac{1}{n\sqrt{n^2-1}} dn$$

$$\text{let } n = \sec \theta$$

$$\frac{dn}{d\theta} = \sec \theta \cdot \tan \theta$$

$$dn = \sec \theta \cdot \tan \theta d\theta$$

$$\text{When } n=1, \theta=0$$

$$\text{When } n=\sqrt{2}, \theta = \frac{\pi}{4}$$

Now,

$$\int_0^{\pi/4} \frac{1}{\sec \theta \sqrt{\sec^2 \theta - 1}} \times \sec \theta \cdot \tan \theta d\theta$$

$$= \int_0^{\pi/4} \frac{1}{\sec \theta \cdot \tan \theta} \times \sec \theta \cdot \tan \theta d\theta$$

$$= \left[\int d\theta \right]_0^{\pi/4}$$

$$= [\theta]_0^{\pi/4}$$

$$= \frac{\pi}{4} - 0$$

$$= \frac{\pi}{4}$$

$$c) \int_0^a \frac{1}{a^2 + x^2} dx$$

$$\text{let } x = a \tan \theta$$

$$\frac{dx}{d\theta} = a \sec^2 \theta$$

$$dx = a \sec^2 \theta d\theta$$

When $x = 0$, $\theta = 0$

When $x = a$, $\theta = \frac{\pi}{4}$

Now

$$\int_0^{\pi/4} \frac{1}{a^2 + a^2 \tan^2 \theta} \times a \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \frac{1}{a^2 \sec^2 \theta} \times a \sec^2 \theta d\theta$$

$$= \int_0^{\pi/4} \frac{1}{a} d\theta$$

$$= \left[\frac{1}{a} d\theta \right]_0^{\pi/4}$$

$$= \frac{1}{a} [d\theta]_0^{\pi/4}$$

$$= \frac{1}{a} [\theta]_0^{\pi/4}$$

$$= \frac{1}{a} \left[\frac{\pi}{4} - 0 \right]$$

$$= \frac{\pi}{4a}$$